HANEEN ELMAHDY

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Immediately available for full-time positions

Professional Summary

Computer Science graduate and MSc student researcher with hands-on experience in machine learning, data analysis, statistical evaluation, and Python-based systems. Contributing to STDF-funded Project No. 48289 under Principal Investigator Yasser Assran, with experience in data cleaning, exploratory data analysis, clustering, model benchmarking, PCA/t-SNE analysis, and technical reporting. Skilled in Python, SQL, Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, Flask, and MS SQL Server, with additional exposure to Excel/VBA and Power BI coursework. Seeking an entry-level Machine Learning Engineer, Data Analyst, or Applied AI role.

Skills

Core ML / Data Libraries: Python, **PyTorch**, NumPy, Pandas, **scikit-learn**, **Matplotlib**, Seaborn, Stable-baselines3.
Web / Backend: **Flask**, FastAPI, SQLAlchemy, MS SQL Server, Firebase, PHP.
Programming Languages: **Python**, **SQL**, C#, C++, Java, JavaScript, HTML/CSS, VBA.
Platforms & Tools: **Git**, Docker, Linux (Bash/SSH), Overleaf (LaTeX), Excel/VBA, PowerPoint/VBA, MS Access, Visual Studio, Unity3D (2D/3D/AR), Android Studio (Java), Photoshop.
Languages: **Arabic** (Native), **English** (stronger in technical reading and writing; actively improving speaking; B2 course level)

RESEARCH & INDUSTRY EXPERIENCE

MSc Student Researcher — STDF-Funded Project No. 48289 **Feb 2023 – Present**
Basic Science Grant Hosted at the British University in Egypt
- Analyzed and compared 21 deep-learning clustering methods on NASA's 4FGL Fermi Gamma-ray catalog using three internal validation metrics: Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Index.
- Used PCA and t-SNE to investigate data distributions and model failure modes, guiding preprocessing and model-selection decisions.
- Authored six structured technical reports summarizing experimental setup, results, failure analysis, and proposed next steps.
- Compared two RL exploration components—BiGAN-based novelty estimation, forward-dynamics models, as intrinsic-reward normalization—to inform Ex-MDPO design.

ElectroPi **Jul 2022 – Feb 2023**
Machine Learning Engineer Intern
- Cleaned, preprocessed, and explored real-world datasets with Pandas and NumPy; translated business requirements into model specifications and technical results.
- Built and stress-tested 8+ ML models for product recommendation and classification, applying Dropout and Early Stopping to address overfitting.
- Developed sentiment-analysis pipelines with LSTM and BERT-based architectures, progressing from a CNN baseline through transformer models.

Education

Benha University, Faculty of Artificial Intelligence **Feb 2020 – Present**
MSc, Computer Science (Expected 2026) *Egypt*
- Reinforcement Learning major, Part-time, (In Progress — Thesis Stage).

Modern Academy for Computer Science and Management Technology **May 2015 – May 2019**
BSc, Computer Science *Egypt*
- Grade: Very Good with Honours (82.98%). Graduation Project: Excellence — 4th Primary Math App.

Advanced Projects

Ex-MDPO — Exploration Mirror Descent Policy Optimization | *Python, PyTorch, RLeXplore, BiGAN, PPO, MDPO* **2026**
- Designed an MDPO-based RL exploration extension after a comparative study of four variants, adding BiGAN-based transition novelty, separate advantage estimation for extrinsic and intrinsic rewards, and adaptive combined state/transition novelty; evaluated it on MuJoCo, Atari, and robotic environments in OpenAI Gym.

Deep Learning Time-Series Clustering — 4FGL Fermi Gamma-ray Catalog | *Python, PyTorch, Pandas, NumPy, Matplotlib, Seaborn* **2026**
- Reviewed, implemented, and benchmarked 21 methods for time-series clustering on NASA's 4FGL Fermi Gamma-ray catalog, using Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Index to identify an approach for gamma-ray source classification.

Explainable Reinforcement Learning for Jet Grooming | *Python, TensorFlow, Stable-Baselines3, DQN, PPO, MDPO, GNNs* **2026**
- Benchmarked DQN, PPO, and MDPO for explainable LHC jet grooming on 50,000 simulated jets; PPO significantly outperformed DQN (3.963 ± 0.041 vs. 2.906 ± 0.006; $p = 0.008$), while decision-tree analysis identified angular separation as DQN's dominant feature.

InventoryMS — Full-Stack Inventory Management System | *Python, Flask, MS SQL Server, SQLAlchemy* **Dec 2025 – May 2026**
- Built and deployed an inventory-management system used daily by a family-owned business, replacing manual spreadsheet tracking for products, sales, purchases, customers, suppliers, expenses, wallets, returns, and stock trends.
- Built dashboard KPIs and sales, purchase, and stock-trend reports to support operational monitoring alongside the transaction-management modules.

LLM-Powered CV Screening / Document Matching Tool | *Python, LangChain, OpenAI API, FAISS, FastAPI* **Aug 2024 – Nov 2024**
- Built an automated CV and document-matching system using preprocessing, tokenization, embeddings, and cosine similarity to support semantic matching beyond keyword overlap.

Self-Play Tic-Tac-Toe Value-Learning Study | *Python, TD Learning, ε-Greedy Policy* **Jan 2023**
- Reimplemented and experimented with Sutton and Barto's self-play Tic-Tac-Toe value-learning example during reinforcement-learning coursework, using an MIT-licensed reference implementation as a guide to understand tabular value learning, ε-greedy exploration, temporal-difference updates, self-play training, and end-to-end agent evaluation before applying the same workflow to additional course examples.

YOLOv8 Object Detection & Pose Estimation | *Python, PyTorch, C#, ONNX Runtime* **Apr 2023**
- Converted YOLOv8 object-detection and Hugging Face pose-estimation pipelines to C# with ONNX Runtime for cross-platform industrial deployment and evaluated the deployment with mAP metrics.

Object Detection Pipeline — CNN, Transformer, and YOLO | *Python, YOLO, Transformers, OpenCV* **Dec 2022**
- Developed an object-detection pipeline progressing from CNN baselines through transformer-based detection to YOLO, comparing when each architecture is appropriate.

Medical & Biological Dataset EDA and Classification | *Python, Pandas, scikit-learn, XGBoost* **Oct 2022**
- Performed EDA on heart-failure and Iris datasets and compared SVC, Random Forest, Gradient Boosting, XGBoost, Decision Tree, K-Means, and DBSCAN; used learning curves and cross-validation to investigate overfitting.

Neural Networks from Scratch | *Python, TensorFlow, NumPy* **Nov 2022**
- Implemented ANN, CNN, and LSTM components and training workflows using Python, TensorFlow, and NumPy, demonstrating an understanding of backpropagation, weight optimization, and training dynamics.

Additional project details are available upon request

Earlier Software & Academic Projects

Regularization and Dimensionality Reduction | *Python* **Jul 2021**
- Calculated accuracy, precision, recall, and F1 score while applying missing-value ratio, variance, and correlation filters, Random Forests/Ensemble Trees, PCA, LDA, backward feature selection, and forward feature selection.

Prime Numbers Application | *C#, Windows Forms* **Oct 2021**
- Developed an application for prime-number detection, prime factorization, Goldbach's conjecture validation, and analysis of palindromic, Germain, safe, and Fermat numbers.

AR Simple | *Unity3D, Vuforia* **Jun 2021**
- Built an augmented-reality application demonstrating basic marker-based AR functionality and interaction mechanics.

4th Primary Math App | *Java, C#, Python, Firebase* **Nov 2018 – May 2019**
- Built an educational mathematics application with Android Studio/Java, Unity3D/C# lessons and mini-games, a Python neural question-generation model via Chaquopy, and Firebase progress tracking.

Image Processing Application | *C#, OpenCV* **Jan 2019**
- Developed a Windows Forms application supporting 3x3 filters, maximum image-level calculations, histograms, and grayscale/RGB channel visualization.

Library Management System | *C#, SQL, MS Access, PHP* **2016 – 2018**
- Built a library system with C#, SQL, and MS Access administration features plus an HTML/CSS/JavaScript/PHP web interface for search, selection, and book purchasing.

Supermarket Application | *JavaScript, HTML, CSS, PHP, MySQL* **Jan 2019**
- Built a web application for shopping-list management, item-price updates, customer lists, and shopping-cart interactions.

Research Manuscripts— Submitted

- Elmahdy, H., Elsawy, A., & Sakr, F. Z. ***Quantum Reinforcement Learning: A Systematic Literature Review.*** International Journal of Applied Intelligent Computing and Informatics. Submitted.
- Elmahdy, H., & Assran, Y. ***Reinforcement Learning in Particle Physics: A Unified Review of Simulation Platforms and Real-World Applications.*** Artificial Intelligence Review. Submitted.
- Elmahdy, H., Assran, Y., Elsawy, A., & Sakr, F. ***Explainable Reinforcement Learning for Jet Grooming.*** Engineering Applications of Artificial Intelligence. Submitted

Self-Directed Learning & Technical Training

- **AI-Powered Assessment Design** — Nile University, Feb 2025.
- **Build with LangChain**, YouTube, In Progress.
- **Mini-RAG from Notebooks to Production**, YouTube, Sep 2024.
- **Reinforcement Learning**, Udacity, 2023.
- **Clean, transform, and load data in Power BI**, Microsoft (online), Jan 2023.
- **Quantum Computing Introduction**, IBM (online), Oct 2022.
- **Machine Learning**, Stanford University (Coursera), Mar 2020 – Sep 2021.